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CRITICAL SUCCESS FACTORS AND MAIN RISKS OF FAILURE IN SOFTWARE DEVELOPMENT PROJECTS

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ABSTRACT

This paper explores the critical success and failure factors in software development projects, aiming to compare the well-established findings in the literature with the current perceptions and experiences of industry professionals. Through a systematic analysis of academic studies and official articles from recent decades, the main people- and process-related variables influencing project outcomes are identified. Afterwards, a targeted questionnaire was distributed to collect empirical data from the IT community, in order to assess any changes in the priorities given to critical factors compared to what emerges from the literature. The comparative analysis of the results highlights similarities and differences in the importance of traditionally relevant factors and the emergence of new elements not yet structured in previous studies. Finally, possible future developments and practical implications for the management and optimization of software development processes in constantly evolving competitive environments are discussed.

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CHAPTER 1.
INTRODUCTION

Software development is now a vital activity for the competitiveness of private companies and public organizations, yet it remains notoriously prone to inconsistent results. Since the early days of project management, the "triple constraint"—time, cost, and quality—has been used as a standard for evaluation: a project is considered successful if it delivers the expected product within the allocated budget and deadline [11]. However, this perspective assumes that goals and requirements are clearly defined from the start, and that the main task of the project manager is to create a detailed plan and keep it under control.

With the evolution of markets, technologies, and business models, this view has proven to be too narrow. Industry studies show that a project can meet deadlines and stay within budget, yet still be considered a failure because it does not generate benefits for the customer or economic value for the organization [12][13]. On the other hand, initiatives that exceed their budget may be seen as "successful" if they produce software that fully meets user needs or open up new market opportunities. In the absence of a shared evaluation framework, the scientific and professional communities continue to question what “project success” truly means.

In this context, the literature introduces the concept of a Critical Success Factor (CSF): any condition whose presence and proper management significantly increase the likelihood of achieving project goals. When these conditions are absent, overlooked, or poorly managed, they become actual causes of failure, leading to deviations from expected cost, time, quality, or stakeholder value that cannot be recovered. Studies generally group CSFs and failure causes into two major and interdependent domains. The first is the people domain, which includes the availability of adequate skills, team motivation, project manager leadership, and clear communication with sponsors and stakeholders. The second concerns processes: realistic planning, scope control, proactive risk management, use of progress metrics, and the selection of a development model appropriate to the level of uncertainty (traditional, agile, or hybrid).

These two domains do not operate in isolation: a flawless process can be undermined by a lack of skills or internal team conflicts; a highly qualified group can fail if requirements change without control or if risk management is inadequate. Hence, the importance of a systemic view in which people and processes reinforce each other: success develops when both pillars work in synergy, while failure occurs when even one of them collapses, spreading instability throughout the project.

Contemporary experience shows that the “triple constraint” measures execution efficiency, but not necessarily effectiveness—that is, the ability of the project to create lasting value for its recipients. Clearly understanding what constitutes a success or failure factor—and how these elements are intertwined in the domains of people and processes—is therefore the first step toward reducing the

high failure rates that, despite decades of methodological progress, still characterize the software industry.

1.1. RESEARCH OBJECTIVES

This study stems from the need to verify whether the critical success and failure factors identified in the literature over recent decades still effectively explain the outcomes of today's software projects, despite the evolving context in which such projects take place. To this end, the research is structured into two complementary phases. The first phase involves a critical review of the main academic and professional publications, aimed at identifying and summarizing the set of people- and process-related factors that the scientific community recognizes as crucial for the success—or, in their absence, the failure—of a software development project. The second phase consists of collecting empirical evidence through a questionnaire addressed to IT professionals (project managers, analysts, developers, and managers) with direct experience in software projects.

The main goal is to compare the results of the literature review with current expert perceptions in order to answer a central question: do the critical factors identified in the past still hold the same relevance, or has the profile of success and failure changed over time? From this question, several specific objectives follow:

- To assess the extent to which the evidence found in the literature matches the current experiences of professionals regarding the relative importance of the various people- and process-related factors;
- To identify any shifts in priority, highlighting factors that were once considered marginal but are now seen as crucial, or conversely, traditionally critical factors whose importance has diminished;
- To detect emerging factors not yet structured in the literature, but reported by professionals as key to the success of projects in the current competitive landscape.

Through this dual path—analytical and empirical—the study aims to validate, integrate, or recalibrate the existing body of knowledge, helping to bridge the gap between theory and practice and offering an up-to-date reference for professionals and researchers working in the field of software project management.

1.2. LITERATURE REVIEW

The analysis of the literature forms the theoretical foundation of this research. Systematically examining the scientific and professional contributions produced over the years allows not only for a better understanding of how the concept of success in software projects has evolved, but also for the identification of the factors considered most relevant in positively or negatively influencing project outcomes.

Despite the widespread use of project management tools, methodologies, and standards, studies report that software projects continue to show high failure rates. For this reason, over the past forty years, numerous authors have attempted to identify the elements that determine the success or failure of such initiatives, with a particular focus on aspects related to people and processes.

This section presents a curated selection of ten key articles, chosen for their academic relevance, methodological rigor, and impact on the scientific debate. The contributions analyzed cover a broad time span, which is useful for highlighting both the stability of some theories and the changes that have emerged over time in response to new organizational and market challenges.

For each article, the reference context, objectives, methodology, and—above all—the main findings concerning critical success factors and causes of failure in software development projects are presented. The aim of this review is twofold: on one hand, to provide a concise and structured overview of the main evidence available in the literature; on the other, to create a point of comparison with the empirical data collected through the questionnaire distributed to industry professionals, in order to assess the persistence, transformation, or emergence of new factors in the current context.

1.2.1. Critical Factors in Successful Project Implementation (1987)

The article "Critical Factors in Successful Project Implementation" [1] by Jeffrey K. Pinto and Dennis P. Slevin, published in February 1987 in IEEE Transactions on Engineering Management (vol. 34, no. 1, pp. 22–27), aims to empirically identify the critical factors that determine project success, to organize them into a coherent process model, and to lay the groundwork for a diagnostic tool that project managers can use. To achieve these objectives, the authors involved fifty-two evening MBA students from the University of Pittsburgh, all of whom were full-time employees with recent experience in project teams. Each participant was asked, using the Project Echo technique, to list five actions that they believed were crucial to ensure the success of a project, defined as "successful" because it brought about organizational change. The 236 actions collected were independently categorized by two experts, who agreed on ten distinct categories; intercoder agreement was considered satisfactory, and after removing duplicates and irrelevant entries, 94 valid actions remained.

The qualitative analysis revealed ten critical success factors: project mission, top management support, planning and scheduling, client consultation, personnel selection and motivation, adequacy of technical skills, final client acceptance, monitoring and feedback systems, effective communication, and troubleshooting ability. The authors organized these factors into a quasi-sequential framework: some (such as mission, support, and consultation) must be emphasized in the early stages, while others (such as communication, monitoring, and troubleshooting) operate across the entire project lifecycle. The resulting model provides empirical confirmation—apart from minor exceptions—of theoretical taxonomies proposed in earlier studies [14][15][16], thus addressing the lack of systematic evidence supporting those hypotheses.

Based on this framework, the authors developed a prototype Likert scale, called the Project Implementation Profile (PIP), which assigns each factor a score from zero to one hundred. This allows the project manager to obtain a quantitative profile of the project, useful during the planning stage, execution, and post-project review to diagnose weaknesses and guide corrective actions. In conclusion, Pinto and Slevin demonstrate that project success depends on a combination of organizational, behavioral, and technical factors, offer a practical model for managing them, and launch a research path grounded in empirical evidence rather than theoretical prescriptions alone.

1.2.2. THE CAUSES OF PROJECT FAILURE (1990)

In their article "The Causes of Project Failure" [2], published in November 1990 in IEEE Transactions on Engineering Management (Vol. 37, No. 4), scholars Jeffrey K. Pinto (University of Maine) and Samuel J. Mantel, Jr. (University of Cincinnati) conduct an empirical investigation into the causes of project failure, with a particular focus on controllable managerial factors that can jeopardize the success of a project. Their work stems from the observation that, despite the growing adoption of project-based approaches, failures still occur frequently and often without a clear understanding of their causes. In particular, the authors emphasize that the very concept of failure is subjective and ambiguous, and that the definitions used to assess project outcomes can significantly influence how results are interpreted.

The main objective of the study is to identify, through a systematic methodology, the critical factors that contribute to the failure of research and development (R&D) projects. In this context, the study also examines whether the influence of such factors varies depending on three contextual variables: (1) the definition of failure adopted, (2) the stage of the project life cycle, and (3) the subjective perception of the project managers involved. The authors adopt a model that distinguishes project failure across three key dimensions: the efficiency of the implementation process (in terms of time, cost, and technical goals), the perceived quality of the outcome, and the final customer's satisfaction. This approach allows success or failure to be considered in a more nuanced and multidimensional way, recognizing that a project can be efficient but pointless, or well-received by the client despite exceeding time or budget limits.

The methodology is based on data collected through a questionnaire sent to a sample of project managers who were members of the Project Management Institute (PMI). Participants were asked to select an R&D project they personally considered a failure and to evaluate it using the Project Implementation Profile (PIP), a tool previously developed by the same authors. The PIP is based on ten critical success factors identified in the literature, including: clear mission, top management support, detailed planning, internal communication, emergency management (troubleshooting), monitoring, client involvement, and adequacy of technical staff. Participants rated the presence of each factor in the selected project using a 5-point Likert scale. In addition, 13 items were used to separately assess the three previously described dimensions of failure.

The results of the statistical analysis, conducted through factor analysis and multiple regression, show that the causes of failure in R&D projects are not uniform but depend on the criteria used for evaluation. When failure is defined in terms of internal efficiency, the most strongly associated factors are the absence of detailed planning and poor emergency management skills. When failure relates to perceived quality, the main cause is the lack of a clear mission from the beginning. Lastly, when client satisfaction is used as the criterion, variables such as user acceptance and the technical adequacy of the delivered solution become more relevant. The study also highlights that failure in R&D projects is often complex and multifactorial, as these initiatives are characterized by higher uncertainty, non-linear iterations, evolving goals, and lower result predictability compared to other types of projects.

One particularly relevant finding of the study concerns the cross-cutting importance of troubleshooting, meaning the ability to manage unexpected events, crises, and deviations from the original plan. This factor appeared as one of the most recurrent in failed R&D projects, regardless of the project phase or evaluation criterion used. Continuous monitoring and active client involvement during the project development also proved to be decisive in avoiding failure. The authors stress that, in R&D environments—where uncertainty is high and solutions are often experimental—it is essential to have flexible control structures, multidisciplinary teams, and a strong focus on the end-use value of the developed product or service.

In conclusion, Pinto and Mantel propose a view of project failure not as a binary event (success vs. failure), but as a multidimensional and context-dependent phenomenon. The results of their study suggest that, in order to reduce the risk of failure in R&D projects, it is necessary to adopt a flexible and targeted management approach that considers the specific characteristics of the project, stakeholder expectations, and the evolutionary stages of development. The study thus offers a solid contribution to both the theory and practice of project management, providing useful tools for analyzing and preventing failure in the most innovative and uncertain project environments.

1.2.3. THE "REAL" SUCCESS FACTORS ON PROJECTS (2002)

In the article "The 'real' success factors on projects" [3], published in 2002 in the International Journal of Project Management (Vol. 20, pp. 185–190), Terry Cooke-Davies presents a critical and empirically grounded reflection on the factors that determine project success. The paper aims to provide a detailed answer to the central question of which factors are truly critical for project success, going beyond the generic responses that have characterized decades of managerial literature. The author emphasizes that, in order to answer this question comprehensively, it is necessary to distinguish among three different questions: "What factors lead to successful project management?", "What factors lead to project success?", and "What factors lead to consistently successful projects within an organization?"

To address these questions, Cooke-Davies adopts an empirical methodology based on data collected through an international network of knowledge-sharing groups. These networks, launched in 1993

by Human Systems Ltd., have over time involved more than 70 multinational and national organizations across Europe, North America, and Australasia. The analysis presented in the article focuses on 136 projects carried out between 1994 and 2000 across 23 different organizations. Data were gathered through detailed questionnaires and structured comparisons between management practices and project performance. The author's work was also developed in parallel with an academic research program that culminated in his PhD dissertation at Leeds Metropolitan University.

The main findings of the research are structured around twelve critical success factors, divided into three groups corresponding to the three levels of analysis. For success in project management (meaning the achievement of time, cost, and quality targets), eight key factors were identified, mainly related to the maturity of risk management processes, clarity of organizational responsibilities, project duration, and strict control of changes and performance baselines. Regarding overall project success, the author identifies one additional critical factor: the presence of an effective process for managing and realizing expected benefits, which requires real collaboration between the project team and operational management. Finally, to ensure systemic and recurring success across projects within the organization, three more factors are highlighted: strategic alignment through portfolio and program management practices, availability of integrated metrics to measure both ongoing performance and post-project success, and a structured organizational learning system based on experience.

The contribution of the article also lies in its clear distinction between "project management success" and "project success," and in its effort to link the success of individual projects to the broader success of the organization, understood as the ability to generate long-term value. Although the paper does not provide a formal definition of project success or failure, it clarifies that management success refers to operational efficiency, while project success concerns strategic effectiveness—and that both are necessary but not sufficient conditions for achieving organizational success.

Finally, although the article does not explicitly list "human" factors among the critical ones, the author underlines that every process is carried out by people and that the effectiveness of each factor depends on the quality of human interaction and leadership. Thus, the human component is an integral—though not separately defined—part of each success factor identified.

1.2.4. Critical Failure Factors in Information System Projects (2002)

In the contribution "Critical Failure Factors in Information System Projects" [4], published in 2002 in the International Journal of Project Management, author K.T. Yeo, a professor at the Division of Systems and Engineering Management at Nanyang Technological University in Singapore, offers an in-depth analysis of the causes leading to the failure of information system (IS) projects. The study stems from the observation that, compared to other high-tech domains, IS projects have a significantly higher failure rate, revealing weaknesses not only in technology but especially in organizational and managerial aspects. The article has two main objectives: on one hand, to propose

a systemic framework capable of representing the complex dynamics that lead to IS project failure; on the other, to use this conceptual structure to classify and critically evaluate the various categories of factors that negatively influence project success.

To achieve these aims, Yeo develops a model called "Triple-S", based on three interrelated systems: the system of strategic planning and project delivery (Sp), the organizational system that receives and uses the information system (S1), and the actual formalized information system itself (S2). The core assumption is that project success or failure does not depend solely on the technology implemented (S2), but also on its integration with the human and organizational context (S1), through a conscious process of planning and management (Sp). The model, inspired by systems theory and particularly by "soft systems" approaches, makes it possible to understand the interaction between technical, cultural, managerial, and political aspects, providing a broader interpretive lens than traditional frameworks.

The methodology used by the author is empirical and based on a survey conducted in November 2000 through a questionnaire distributed to a sample of professionals involved in IS projects that were considered problematic or failed. A total of 92 valid responses were collected, each referring to a single failed project based on the respondent's direct experience. The questionnaire asked participants to evaluate the perceived importance of a series of critical factors, selected based on an in-depth review of the literature. The collected data were then analyzed using average Likert scale scores, in order to identify the most frequently mentioned and significant areas of influence.

The findings of the study confirm that the failure of an IS project is rarely due to a single technical issue. On the contrary, it clearly emerges that the main criticalities lie in the domain of project planning and management (Sp), as well as in the organizational and cultural context in which the project is embedded (S1). Among the most frequently cited factors are the underestimation of time and resources, weak requirements definition, lack of risk analysis, poor user involvement, and a corporate culture resistant to change. Technical problems or issues related to technology choice (S2) were mentioned much less frequently. In particular, the statistical analysis showed that the areas with the highest average scores were project planning, corporate culture, and managerial control.

This article serves as a key reference in the study of information system failures and emphasizes the need for project managers to adopt systemic, participative, and reflective approaches when managing complex projects.

1.2.5. Defining Success for Software Projects: An exploratory Revelation (2006)

In the contribution titled "Defining Success for Software Projects: An Exploratory Revelation" [5], published in 2006 and funded by the Indian Institute of Management Indore, authors Nitin Agarwal and Urvashi Rathod systematically and thoroughly investigate the concept of success in software projects, aiming to understand which criteria are actually used by industry professionals to define it. Unlike traditional literature, which tends to evaluate project performance based on adherence to

predefined time, cost, and quality targets, this study starts from the assumption that such indicators may not be shared by all stakeholders, especially between internal project team members and external stakeholders.

The main objective of the article is to explore which characteristics make a software project "successful" according to the professionals working on it. Specifically, the study seeks to answer two key questions: (1) which project dimensions are perceived as most relevant in defining success by software professionals, and (2) whether these perceptions vary according to professional role, by comparing the opinions of programmers, project managers, and customer account managers. The authors argue that existing literature often overlooks the internal perspective of software projects, and that building a more robust evaluation framework requires starting precisely from this viewpoint.

The methodology used is based on an exploratory quantitative study conducted through a structured questionnaire. The sample, selected intentionally, consists of 105 software professionals from 12 Indian companies with a CMM level of 3 or higher, operating in various sectors (telecommunications, finance, manufacturing, etc.) and involved in both custom software development and product creation. Participants were grouped into three professional categories: programmers, project managers, and account managers. Each respondent was asked to share their opinion on three different scenarios: (1) a project urgency situation where one must prioritize between time, cost, or functionality/quality (scope); (2) the identification of a single most important criterion for defining project success; and (3) the ranking of four criteria—time, cost, functionality, and quality. The data were statistically analyzed using the Chi-square test for frequency analysis and Kendall's coefficient of concordance (W) for rankings, while open-ended responses were examined qualitatively through thematic categorization.

The results of the study are surprisingly consistent. In all three proposed scenarios, the criterion of "scope"—including functionality and product quality—is considered the most important, regardless of the respondent's role. Even in situations of project urgency, most professionals prefer to delay delivery or increase costs rather than reduce software functionality. Functionality is consistently considered more important than internal quality, and far more important than time or cost. Cost, in particular, emerged as the least prioritized criterion in all analyses. Only a small percentage of participants spontaneously referred to alternative criteria such as customer satisfaction or the strategic value of the project, suggesting that the internal organizational culture is strongly focused on the completeness and functional coherence of the software product.

In summary, the study confirms the hypothesis that perceptions of success in software projects are strongly influenced by the operational role held, but it also shows significant agreement among professionals with different responsibilities. The results challenge the systematic use of evaluation models based solely on time and budget and invite a rethinking of the concept of project success, recognizing the central role of expected functionality and perceived quality of the final product. The authors conclude by encouraging future research to further explore the link between achieving

scope and customer satisfaction, with the goal of developing a more comprehensive, realistic, and widely shared model for evaluating project success.

1.2.6. THE INFLUENCE OF PROJECT MANAGERS ON PROJECT SUCCESS CRITERIA AND PROJECT SUCCESS BY TYPE OF PROJECT (2007)

The study "The Influence of Project Managers on Project Success Criteria and Project Success by Type of Project" [6], conducted by Ralf Müller (Umeå University, Sweden) and Rodney Turner (Lille School of Management, France), was published in 2007 in the European Management Journal (Vol. 25, No. 4, pp. 298–309). This work is part of a growing body of research recognizing the need to tailor the project management approach based on project type, industry context, and the individual characteristics of project managers.

The central goal of the research was to explore whether, and how, the importance attributed to project success criteria and the levels of success actually achieved vary according to different dimensions. Specifically, the authors posed the following research questions: (1) whether different success criteria are relevant for different types of projects or industries; (2) whether different types of projects perform differently with respect to those criteria; (3) whether project managers assign different levels of importance to success criteria based on their personal characteristics; (4) whether their performance varies based on these traits; and (5) whether there is a relationship between the importance assigned to certain criteria and the perceived success of the project in those same areas.

To investigate these questions, Müller and Turner used a quantitative methodology based on a global web survey that collected 959 complete responses. The questionnaire was divided into five sections: project classification (according to six attribute areas defined by Crawford et al., 2005), evaluation of the importance of seven success criteria, assessment of perceived success across ten criteria, presence of project management certification, and collection of demographic data (gender, age, nationality, education level, job title, industry sector). Data analysis included ANOVA tests, regression analysis, and subgroup comparisons (e.g., between high- and low-performance projects).

The results showed that the importance attributed to success criteria varies significantly depending on project complexity, contract type, and industry sector. In highly complex projects, project managers tend to emphasize the satisfaction of customers, suppliers, and stakeholders. Recurring business is more important in the private sector than in the public sector. Moreover, in high-performance projects, there is a stronger focus on all stakeholders involved. However, overall perceived project success does not vary significantly based on project type or industry, although some differences do emerge when isolating high-performance projects. For instance, ICT and organizational change projects tend to be more successful than engineering projects, and projects managed in the project manager's home country achieve better results than those led by expatriates.

Regarding project manager characteristics, age and nationality influence how importance is assigned to success criteria: older managers place more value on team satisfaction, while European

managers generally assign less importance to end-user, supplier, and stakeholder satisfaction compared to their counterparts from other regions. Additionally, in high-performance projects, certified project managers achieve better outcomes, especially in meeting user requirements. However, certification alone is not enough: managers with poor performance remain so even if certified, while certification mainly benefits those who already have strong skills.

One of the most significant theoretical contributions of the study is the demonstrated relationship between the importance attributed to success criteria and the actual success achieved. The data show that placing greater importance on specific criteria—especially team, end-user, and customer satisfaction—significantly increases the likelihood of achieving positive results in those same areas. This confirms the hypothesis that the managerial focus of the project manager has a direct impact on perceived performance.

The study also offers important practical implications for sponsors and organizations. Among them: project manager assignments should consider project complexity and manager experience to avoid underutilization; local managers are preferable, especially in culturally sensitive contexts; the most effective project managers are those who follow the entire project lifecycle, from feasibility to closure; and finally, practical experience combined with certification is the best indicator for identifying high-performing project managers.

1.2.7. WHY DID YOUR PROJECT FAIL? (2009)

The article "Why Did Your Project Fail?" [7] by Narciso Cerpa and June M. Verner, published in December 2009 in Communications of the ACM (Vol. 52, No. 12), addresses the issue of failure in software development projects by seeking to understand the most common causes through an empirical investigation. The authors note that, despite the experience accumulated in the sector since the 1960s, software projects continue to fail at an alarming rate, with little sign of improvement over time. Previous studies—such as those by the Standish Group [11]—have frequently pointed out that many technology initiatives suffer from budget overruns, significant delays, and general user dissatisfaction. However, much of this research has focused on a few high-profile cases and has rarely approached the issue in a systematic and quantitative way. This study therefore aims to update and broaden the understanding of failure factors by examining a larger and more representative sample of projects, including common experiences often neglected in traditional literature.

The article has three main goals. First, to identify the development and management practices most frequently associated with failure. Second, to provide large-scale empirical evidence to confirm or refute previously identified failure factors. Third, to offer concrete recommendations to help prevent similar mistakes in future projects. The focus is not only on the technical management of the project, but also on organizational dynamics, stakeholder relationships, and the human dimension of development work.

To conduct the study, the authors designed and administered a structured questionnaire consisting of 88 questions, grouped into seven thematic areas: sponsor/executive leadership, clients/users, requirements, estimation and planning, development process, project management, and development team. The questionnaire was developed after discussions with over 90 professionals in the field, ensuring the practical relevance of the questions. The survey was first distributed to a large group of developers at a U.S. financial institution, and then to three additional groups of professionals working in companies located in the northeastern United States, Australia, and Chile. Each participant was asked to provide responses based on two different projects: one they considered a success and one they considered a failure, according to their personal experience. Thus, the classification of success or failure was subjective, reflecting the professional judgment of those who participated in the study.

In total, data were collected on 304 software projects. After removing entries with significant missing data, the final sample included 235 projects, of which 70 were classified as failures: 49 were internal projects, and 21 were outsourced. These were not exceptional or publicly known cases, but ordinary projects managed within typical companies. About 60% of them had teams of fewer than 10 developers, while only 10% had more than 55. The data were analyzed using frequency analysis and correlation techniques to identify the most relevant patterns.

The findings are clear: software projects do not fail due to a single cause but rather due to the interaction of multiple, often interconnected, factors. The four most frequent factors found in failed projects were: an imposed delivery date that hindered the normal development process; a significant underestimation of the required resources and time; a lack of risk reassessment or management throughout the project life cycle; and the absence of incentives or recognition for staff working overtime. Additional critical issues included a lack of user involvement in scheduling, inappropriate development methodologies, insufficient structured reviews at the end of each phase, and unrealistic client expectations. The authors also highlight factors related to team motivation, such as unpleasant work experiences, excessive workloads, and negative impacts on personal life.

The comparison between internal and outsourced projects is also noteworthy: internal projects were more likely to suffer from poor requirements quality and inadequate personnel management, while outsourced projects faced greater challenges in risk management and change control. However, both types shared common weaknesses in planning and timeline adherence.

The research reaffirms the importance of involving the project manager in estimation activities, adopting suitable methodologies, and recognizing the impact of organizational decisions on team motivation and the overall success of the project.

1.2.8. Critical Success Factors Assessment in Software Projects (2013)

The article "Critical Success Factors Assessment in Software Projects" [8], published in 2013 and presented at the Science and Information Conference (October 7–9, London, UK), was written by

Rabia Hashim and Dr. Muhammad Abbas (both from the Department of Computer Engineering, National University of Sciences and Technology, Islamabad, Pakistan), and Muhammad Hashim (Benazir Income Support Programme, Islamabad). The study contributes to the body of research focused on risk management in software projects, with particular emphasis on identifying the Critical Success Factors (CSFs) that influence the outcome of IT initiatives.

The main objective of the article is to compare IT project performance and identify those critical factors whose presence or poor management can lead to project failure. The authors point out that risks in software projects typically stem from unclear requirements, uncontrolled scope changes, overly optimistic planning, budget overruns, poor communication, and weak change management. For this reason, the article aims to provide a practical contribution to the understanding of such risks and to support decision-making by project managers, in order to mitigate negative outcomes and improve project success rates.

The methodology employed consists of two complementary phases. First, a systematic literature review was conducted to identify existing approaches for evaluating software project risks published since 2000. This review allowed the authors to select and analyze relevant studies that addressed specific research questions such as: What are the most critical risks in software projects? What is the impact of risk mitigation tools? What role do organizational policies play? In the second phase, the authors conducted an online survey targeted at software professionals with at least five years of experience, working in both national and multinational companies. The questionnaire was sent to 47 organizations and received 40 responses from senior roles including general directors, presidents, and project managers.

The results highlight that scope change is the most critical risk factor, followed by unrealistic requirements, unclear specifications, optimistic planning, and communication problems. Moreover, the use of risk assessment tools was found to significantly enhance a team's ability to identify risks at an early stage. Planned mitigation activities, leadership involvement, and the presence of strong organizational policies were also shown to be crucial for project success. Survey participants confirmed that organizations with structured policies and a proactive approach to risk management achieve better project performance. Another important contribution of the study is the identification of an inverse relationship between risk level and project performance, revealing that complexity and project size—especially in medium-scale projects—can increase vulnerability.

In conclusion, the article emphasizes that effective management of critical success factors requires an integrated approach based on early identification, active mitigation, and structured organizational governance. As a direction for future work, the authors propose the development of an automated predictive tool capable of identifying risks in software projects across four key dimensions: people, processes, product, and technology. This study is particularly useful for researchers and professionals involved in risk management for software projects, offering concrete insights into the factors that influence project success or failure.

1.2.9. A Correlational Study on Project Management Methodology and Project Success (2019)

The study "A Correlational Study on Project Management Methodology and Project Success" [9], conducted by Michael Pace and published in 2019 in the Journal of Engineering, Project, and Production Management (Vol. 9, No. 2, pp. 56–65), aims to systematically investigate the relationship between the project management methodology used and reported project success. The research stems from the observation that, despite the wide availability of tools, techniques, and methodologies for project managers, project failure rates remain high and consistent over time. Previous studies, such as those by Budzier and Flyvbjerg [12] and Serrador and Pinto [13], revealed conflicting results regarding the effectiveness of traditional versus agile methodologies, thus paving the way for further empirical exploration.

The main objective of the study is to explore whether, and to what extent, the methodology adopted (agile, traditional, or hybrid) can influence project success as perceived and reported by North American project managers with at least five years of professional experience. Additionally, the research considers two potential moderating variables: the industry sector (technology vs. non-technology) and the project manager's level of experience.

To achieve this objective, the author employed a non-experimental, correlational quantitative research method. The sample was selected using simple random sampling from a panel of PMP-certified project managers residing in North America. Participants were required to be at least 25 years old and familiar with multiple project management methodologies. The data collection instrument was an online questionnaire based on the work of Serrador and Pinto [13], consisting of 58 questions divided into three sections: demographic information, a description of a successful project, and a description of a less successful project. Responses were collected using a 5-point Likert scale. A total of 367 valid responses were obtained. Data were analyzed using SPSS software, with Pearson correlation and linear regression applied to assess the strength and significance of relationships between variables.

The results showed that project management methodology has only a weak correlation with reported project success, particularly in non-technology sectors. However, in technology-related projects, the correlation between the adopted methodology and perceived success was moderate and statistically significant, with a more positive trend towards agile methods. Furthermore, the analysis revealed that project manager experience does not significantly moderate the relationship between methodology and project success in any of the models tested. These findings challenge the assumption that simply adopting a specific methodology can consistently improve project outcomes, and suggest that other factors—such as the organizational context, the maturity level of practices, and the degree of methodological tailoring—may be more decisive.

The author concludes by emphasizing the importance of greater methodological flexibility in project management, suggesting that more dynamic and adaptive decision-making is essential in volatile, uncertain, complex, and ambiguous (VUCA) environments.

1.2.10. Agile versus Waterfall Project Management: Decision Model for Selecting the Appropriate Approach to a Project (2021)

The article "Agile versus Waterfall Project Management: Decision Model for Selecting the Appropriate Approach to a Project" [10], written by Theo Thesing, Carsten Feldmann, and Martin Burchardt, was published in 2021 in Procedia Computer Science. This contribution is part of the ongoing debate on contemporary project management, addressing the practical challenge of selecting the most suitable methodological approach for managing a project, specifically by comparing the two dominant paradigms: the traditional waterfall approach and the agile approach.

The article has a dual purpose. On one hand, the authors aim to understand, through the direct experience of professionals, the perceived differences, benefits, and limitations of agile compared to the classical model. On the other hand, they seek to develop a structured decision-making model that can help project and program managers consciously choose the most appropriate procedural model for a specific project. The article thus attempts to fill both a practical and theoretical gap: despite the extensive existing literature on management methods and techniques, a simple, empirically grounded, and widely applicable model for selecting a method is still lacking.

Methodologically, the study is based on an exploratory qualitative research design, structured in two main phases. In the first phase, the authors conduct a systematic literature review across six academic databases, identifying 52 relevant sources. This review highlights the absence of a decision model that is both theoretically sound and validated by real-world experience.

In the second phase, the authors conduct 15 semi-structured interviews with project management experts from various sectors (ICT, automotive, finance, engineering, public administration), with diverse professional backgrounds in terms of age, experience, and role (including external consultants). The interviews are analyzed through qualitative content analysis. The resulting insights are then integrated into a modeling process, following principles of model quality such as correctness, relevance, clarity, efficiency, and consistency.

The main findings of the empirical investigation show that professionals perceive significant differences between the two approaches, particularly in planning (rigid and upfront in waterfall, flexible and iterative in agile), work culture, leadership style, and the centrality of process versus plan. The classical approach is valued for its clear roles, stable planning, predictable timing and costs, and efficiency due to standardization. Agile, instead, is considered superior in its adaptability, quick response to errors and changes, risk reduction, and team motivation.

The disadvantages identified for waterfall include the heavy upfront planning effort and lack of flexibility during the project. For agile, the challenges mainly concern its dependence on a

supportive organizational culture, the need for highly autonomous and dedicated teams, and potentially higher costs due to continuous iteration.

The study's most significant theoretical contribution is the development of a two-phase decision model: the first phase involves identifying exclusion criteria that make agile either unfeasible or ineffective (e.g., legal requirements, lack of project modularity, rigid organizational environment); the second phase proposes a detailed evaluation grid based on 15 criteria, grouped into five key categories: project scope, time, cost, organizational context, and team characteristics. Each criterion is scored on a scale from 0 to 4, and the final score guides the selection toward a classical, agile, or hybrid model.

The authors conclude that there is no universally valid methodological model for all projects, and that it is essential to consciously select the approach based on the specific context.

1.3. SCIENTIFIC EVIDENCE AND TECHNICAL INSIGHTS FROM THE LITERATURE

The analysis of major scientific contributions over the past forty years on project management applied to software development clearly reveals a fundamental insight: project success or failure never depends on a single cause, but rather on a complex and interconnected set of factors operating at different levels and stages of the project life cycle.

The literature consistently points to two closely related dimensions: people and processes. Effective management of these two areas creates favorable conditions for successful outcomes, while neglect or weak governance is often at the root of most failures observed.

The Critical Success Factors (CSFs) identified thus represent strategic levers for increasing the likelihood of achieving project goals, while their absence or poor management frequently recurs among the causes of failure. Below are the main CSFs and failure factors identified in the literature, each accompanied by a brief explanation.

CRITICAL SUCCESS FACTORS (CSFS)

1. **Clear project mission**
   Defining goals, purposes, and success criteria from the start in a shared and unambiguous way helps guide the project in a stable and consistent direction.
2. **Top management support**
   Leadership endorsement, in terms of resources, visibility, and legitimacy, ensures continuity and authority for the project initiative.
3. **Realistic planning and scheduling**
   Accurate planning based on real data helps reduce exposure to structural and operational risks.

4. **Strict change and baseline control**
   Keeping deviations in scope, time, and objectives under control is essential to avoid overruns or misalignment.
5. **Client and end-user consultation and involvement**
   Ongoing dialogue with stakeholders allows for timely feedback, clarified expectations, and improved product adoption.
6. **Personnel selection and motivation**
   A well-chosen, motivated, and recognized team works more effectively, cohesively, and resiliently.
7. **Adequate team skills and leadership**
   Technical and interpersonal skills, along with strong project leadership, are key enablers for managing complexity.
8. **Monitoring and feedback systems**
   Using tracking tools and corrective mechanisms promotes responsiveness and execution quality.
9. **Troubleshooting capability**
   Effectively handling unexpected issues is crucial, especially in dynamic or uncertain environments.
10. **Sufficient and appropriate resources (human, technological, economic)**
    Having qualified staff, suitable tools, and a realistic budget is essential to ensure project execution.
11. **Process maturity and risk management**
    Organizations with established processes and structured risk systems show greater stability and resilience.
12. **Clear organizational responsibilities**
    Defining roles and responsibilities clearly prevents overlap, ambiguity, and operational conflicts.
13. **Team, end-user, and customer satisfaction**
    Projects that end with high satisfaction among creators and users are more sustainable over time.
14. **Planned mitigation and strong organizational culture**
    Proactive risk management supported by solid policies acts as a strong defense against failure.
15. **Methodological flexibility**
    Adopting agile or hybrid approaches when appropriate allows rapid adaptation and improved quality and engagement.

FAILURE FACTORS

1. **Underestimation of time, cost, and resources**
   Overly optimistic planning leads to unrealistic pressure, delays, and overruns.

2. **Undefined or constantly changing scope (scope creep)**
   Lack of a clear project boundary and control system undermines project governance.
3. **Lack of risk analysis**
   Failure to anticipate problems exposes the project to unmanaged critical events.
4. **Weak or inconsistent planning**
   Poorly structured or unrealistic plans create an unstable foundation for the project.
5. **Company culture resistant to change**
   Organizations closed to innovation hinder the success of new initiatives.
6. **Undersized or inexperienced team**
   Small or underqualified teams struggle to cope with complexity, leading to delays and poor quality.
7. **Negative work experiences and personal impact**
   Excessive workload and lack of work-life balance affect team stability and increase turnover.
8. **Inadequate communication**
   Disorganized or incomplete exchanges lead to misunderstandings and decision-making issues.
9. **Lack of user involvement**
   Ignoring user input results in irrelevant or unusable solutions.
10. **Inappropriate methodologies**
    Rigid or context-mismatched approaches reduce responsiveness and project effectiveness.
11. **Absence of structured reviews**
    Lack of interim evaluations prevents learning from mistakes and timely course corrections.
12. **Deadline imposition without analysis**
    Arbitrary deadlines jeopardize development quality and reliability.

In software project management, two methodological approaches have emerged as leading references: Waterfall and Agile [9][10]. These models differ in logic, structure, and change/complexity management and were examined in several of the reviewed studies.

The Waterfall approach features upfront and rigid planning: the phases of analysis, design, implementation, testing, and release follow a linear sequence with limited revision ability. It suits stable, well-defined contexts with high predictability. However, this structure struggles with mid-course changes and is less suited to dynamic or innovative environments.

In contrast, the Agile paradigm is based on short, incremental iterations (sprints), evolutionary planning, and continuous customer involvement. Methods like Scrum, Kanban, and Extreme Programming (XP) promote frequent feedback, adaptability, and team empowerment. Agile delivers value progressively and treats change as inherent to the process, making it ideal for complex, uncertain, or fast-evolving projects.

Across the ten articles reviewed, methodology is not always the central focus but is often cited as a success or failure factor. Specifically:

- Pace (2019) [9] finds a moderately significant correlation between agile use and perceived success in technology sectors;
- Thesing et al. (2021) [10] propose a decision model for choosing between Agile and Waterfall, emphasizing that no single approach is universally superior and that methodology should be selected based on project context, structure, and organizational maturity;
- Other contributions (Cerpa & Verner [7], Yeo [4], Pinto & Mantel [2]) indirectly highlight that using unsuitable methodologies — such as applying Waterfall in highly variable environments — is a common cause of failure.

In conclusion, the literature suggests that methodological flexibility — the ability to adapt the management approach to the specific project context — is now considered a critical competence for success. Effectiveness lies not in strict adherence to a model, but in the ability to interpret project needs and translate them into coherent, sustainable, and value-focused practices.

CHAPTER 2.
MATERIALS AND METHODS

2.1. METHODOLOGICAL APPROACH

This research adopts a combined qualitative and quantitative methodological approach, based on the use of a structured questionnaire as the primary tool for data collection. This choice was driven by the need to obtain objective and comparable numerical feedback, in line with the study's main objective: to verify whether the critical success and failure factors identified in academic literature are still perceived as relevant by professionals working in the IT sector today.

The use of a questionnaire proved particularly suitable for several reasons. Firstly, it allowed for the rapid and broad distribution of the survey, reaching a large number of respondents through digital channels in a limited timeframe. Additionally, its closed and standardized structure ensured consistency in responses, facilitating statistical analysis and the comparison of results with the theoretical data gathered from the literature review.

This methodology is also widely validated in the literature reviewed, where several key studies have employed the questionnaire as their main empirical investigation technique. Notable examples include the works of Pinto and Mantel [2], Cooke-Davies [3], Yeo [4], Agarwal and Rathod [5], Müller and Turner [6], Cerpa and Verner [7], Hashim et al. [8], as well as Michael Pace [9]. These contributions demonstrate that the questionnaire is a well-established and effective tool for exploring opinions, experiences, and critical factors in this field.

The next section will describe in detail the design and structure of the questionnaire, with reference to the choices made during the definition of its content, sections, and administration methods.

2.2. DESIGN AND STRUCTURE OF THE QUESTIONNAIRE

The questionnaire used for this research was created using the Google Forms platform, chosen for its accessibility, ease of use, and compatibility with different devices. The form was distributed via a direct link, made available for several weeks, and shared among IT professionals, with the aim of reaching a broad and varied sample of respondents. The total number of valid responses collected was 93, achieved thanks to extensive sharing and active collaboration with professionals in various roles and organizational settings.

The questionnaire was designed interactively, using filter questions to customize the question flow based on the answers provided. This approach allowed for more targeted content, avoiding irrelevant questions for specific subgroups and thus improving the overall quality of the data collected.

Structurally, the questionnaire was divided into several thematic sections, following a logical progression to guide respondents from general to specific topics:

- **Introductory section:** included socio-demographic questions such as age, education level, and professional role;
- **Project background section:** aimed to gather information on the respondent's involvement in software projects, their level of experience, company type, business area, and the methodologies adopted within the organization;
- **Perceived factors section:** included multiple-choice and ranking questions to identify, according to the participants, the main critical success and failure factors in software projects based on their personal views and general perceptions;
- **Most recent completed project section:** asked respondents to reflect on their most recently completed software project and evaluate it in terms of success or failure, specifying which factors (positive or negative) were actually present during the project's life cycle. This part was crucial for comparing general opinions with concrete experience.

The modular structure of the questionnaire allowed for a comprehensive and coherent exploration of the study's dimensions, while also ensuring a simple and guided experience for participants.

CHAPTER 3.
RESULTS

3.1. DISCUSSION OF RESULTS

The responses collected from the 93 fully completed questionnaires were analyzed. The age of respondents was mostly between 41 and 60 years old (almost 50%), followed by the 26 to 40 age range (36%), as shown in the following Figure 1.

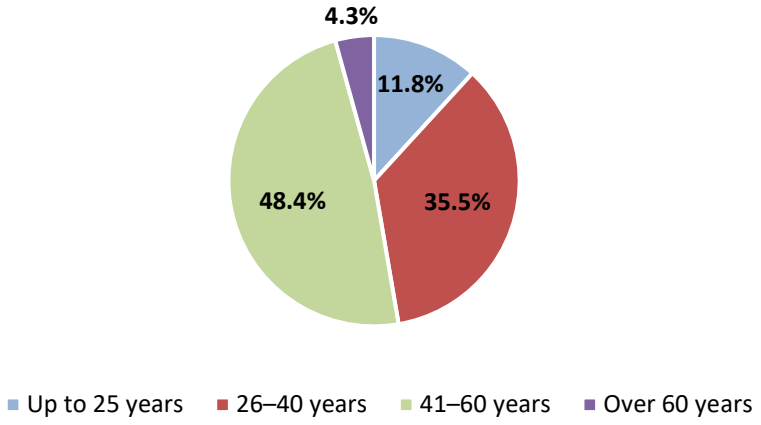

Figure 1 - Age (in completed years)

The gender distribution was nearly balanced: 55% male and 45% female. Regarding education, around 87% of respondents held a university degree (34.4% master's, 30.1% bachelor's, and 22.6% single-cycle degree), while the remaining 12.9% held a high school diploma. The following Figure 2 shows these results.

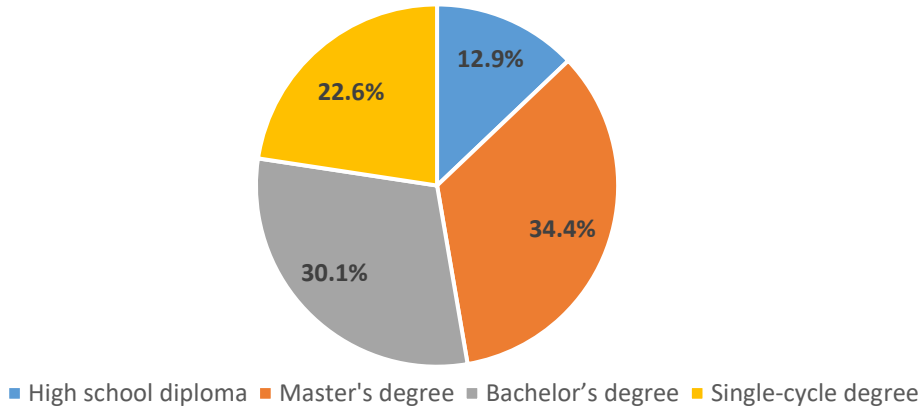

Figure 2 - Educational Qualification

From a professional standpoint, 44% of respondents have worked in the software project field for over 10 years, 20% between 5 and 10 years, and the remaining 36% for less than 4 years. Similarly, 44% had participated in more than 10 projects, 20% in 5 to 10 projects, and 36% in fewer than 5. The following Figure 3 illustrates the distribution of work experience in the field.

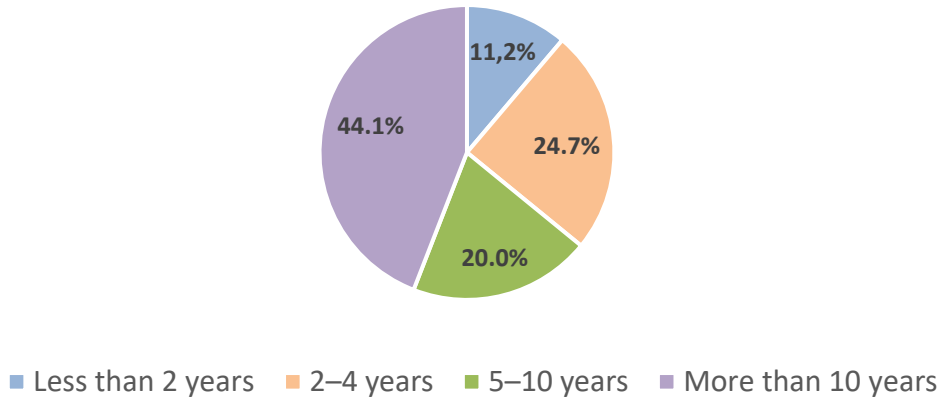

Figure 3 - How many years have you worked in the field of software development and/or project management?

For the most recent project they participated in, respondents were asked to specify their role: 48% held managerial or leadership roles, while the remaining 52% held non-managerial positions, as shown in the following Figure 4.

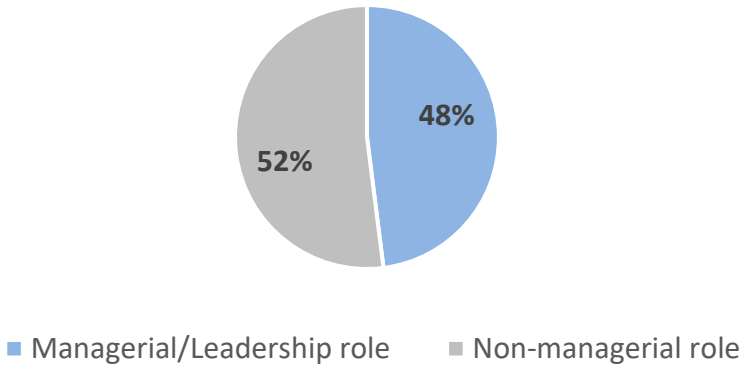

Figure 4 - Role Held in the Most Recent Project

The data analysis therefore also aimed to highlight similarities and differences in perception between managerial and non-managerial figures.

Since the early days of project management, the "triple constraint"—time, cost, and quality—has been used as the standard measure of project performance [11] and remains a widely discussed topic. Respondents were asked to rank these three factors by perceived importance in a software project, from first to third. For the analysis, each position was assigned a score (3 points for first

place, 2 for second, and 1 for third) to allow for quantitative comparison. The results were analyzed separately for managerial and non-managerial respondents. Both groups ranked Quality as the most important factor. However, there was a clear divergence in the next two positions: managers ranked Cost second and Time last, while non-managers reversed this order, giving more importance to Time than to Cost.

The questionnaire then explored the topic of methodologies and approaches. Respondents were asked which methodology they would prefer to use in future projects. The results (summarized in the following Table 1) show that neither group favors a single methodology; instead, both emphasize the need for flexibility and adaptability based on context. Traditional (Waterfall) methodology is almost entirely rejected for future use, with a clear preference toward Agile approaches:

Preferred Methodology to Use in the Future					
Category	**Agile**	**Waterfall**	**Hybrid**	**Depends on the context**	**No preference**
PM	20%	0%	20%	60%	0%
No PM	21%	4%	24%	43%	8%

Table 1 - Preferred Methodology to Use in the Future

Participants were also asked whether, in their opinion, the adopted methodology significantly influences the success or failure of a project. The results (see Table 2 below) indicate that both groups believe methodology choice does have a notable impact:

Methodology influences project outcome		
Category	**Yes**	**No**
PM	84%	16%
No PM	90%	10%

Table 2 - Methodology influences project outcome

The central section of the questionnaire aimed to identify which factors respondents considered most decisive in determining the success or failure of a project. Participants were asked to select the four elements they considered essential for project success. The results are shown below, divided by role.

CRITICAL SUCCESS FACTORS

According to managerial figures, the most relevant success factors are:

- Clearly defined, shared, and stable requirements over time (79%);
- Team competence (77%);
- Customer involvement throughout the entire project lifecycle (72%);
- Realistic planning aligned with available resources (70%).

The factors perceived as least influential are:

- Top management support (10%);
- Presence of iterative feedback (11%).

For non-managerial figures, the most important success factors are:

- Clearly defined, shared, and stable requirements over time (85%);
- Realistic planning aligned with available resources (81%);
- Team competence (65%);
- Sufficient resources (human, technological, financial) (65%).

The least impactful factor for this group is:

- Top management support (4%).

FAILURE FACTORS

According to managerial figures, the main failure factors are:

- Unclear or constantly changing initial requirements (88%);
- Undersized or inexperienced team (79%);
- Unrealistic deadlines (75%);
- Poor communication within the team (70%).

The least influential:

- Poor budget management (2%).

For non-managerial figures, the most impactful failure factors are:

- Unrealistic deadlines (84%);
- Poor communication within the team (80%);
- Unclear or constantly changing initial requirements (77%);
- Undersized or inexperienced team (75%).

The least relevant:

- Conflicts between stakeholders (10%);

In the next section of the questionnaire, respondents were asked to base their answers on the most recent project they had participated in. Of the 93 respondents, 84 stated that they had completed their latest project, while the remainder were still working on it. Only the data from those who completed their project are considered in the following analysis.

Among these 84 respondents: 82% reported that the project was successfully completed, while 18% considered it a failure. The results are presented in the Table 3 below, divided between managerial and non-managerial roles:

Last Project Result		
Category	**Success**	**Failure**
PM	90%	10%
No PM	74%	26%

Table 3 - Last Project Result

After indicating the final outcome (success or failure) of their most recent completed project, respondents were asked to rate, on a scale from 1 (low impact) to 5 (high impact), the extent to which each factor—previously identified among the most critical—had influenced the success or failure of the project.

For managerial and/or leadership figures whose project was successfully completed, all factors with an average score above 4 were selected. These are, in order of score:

- Clearly defined, shared, and stable requirements over time;
- Good cohesion and collaboration among team members;
- Team skills;
- Sufficient resources (human, technological, financial);
- Realistic planning consistent with available resources..

For projects that ended in failure, the most influential factors, again with an average score above 4, were:

- Insufficient available resources;
- Unrealistic customer expectations;
- Unrealistic deadlines;
- Initial requirements that were unclear or constantly changing;
- Undersized or inexperienced team.

From the perspective of non-managerial and/or non-leadership figures, applying the same selection criteria, the most influential success factors were:

- Good cohesion and collaboration among team members;
- Team skills;
- Realistic planning consistent with available resources;
- Clearly defined, shared, and stable requirements over time;
- Sufficient resources (human, technological, financial).

As for projects that ended in failure, the most influential factors (with average scores above 4) identified by non-managerial figures were:

- Undersized or inexperienced team;
- Unrealistic deadlines;
- Initial requirements that were unclear or constantly changing;
- Methodology not suitable for the context;
- Unrealistic customer expectations.

Finally, in the last section, all respondents—both those who declared a successful outcome and those who indicated a failure—were invited to express a judgment on the degree of fulfillment or satisfaction of certain project elements, assigning a score from 1 (not at all) to 5 (completely).

For managerial figures who declared their project successful, the elements with the highest average scores were:

- Overall user satisfaction;
- Fulfillment of functional requirements;
- Fulfillment of technical specifications;
- Product actually in use.

These same four elements were also the most fulfilled in failed projects, but with lower scores, indicating that although they are considered fundamental, they were achieved to a lesser extent in cases of failure.

On the other hand, for non-managerial figures who reported a successful project, the top-rated factors were:

- Fulfillment of functional requirements;
- Fulfillment of technical specifications;
- Product actually in use;
- Overall user satisfaction.

As in the previous case, these four factors were also indicated in failed projects, but with lower average scores, confirming both their perceived importance and, at the same time, their lower realization in unsuccessful projects.

3.2. COMPARISON WITH THE REVIEWED LITERATURE

At this point, after analyzing the results that emerged from the questionnaires, it is possible to proceed with a comparison against the findings presented in the literature reviewed in par. 1.2

TIME / COST / QUALITY

A key element that emerged from the analysis of the so-called "iron triangle"—quality, cost, and time—is the central importance attributed to product quality. Both managerial (PM) and non-

managerial figures indicated quality as the most important factor in a software project, surpassing time and cost constraints in relevance.

This result confirms what was already highlighted in the study by Agarwal and Rathod [5]. The authors point out that, in the context of software development, professionals are generally willing to accept increased costs or extended delivery times in order not to compromise the quality of the final product. This mindset reflects a mature vision of project success—one that goes beyond simply meeting contractual constraints to include user satisfaction, product robustness, and long-term sustainability. The fact that this approach is shared across both managerial and operational profiles strengthens the idea that quality is perceived as an essential foundation of every successful software project.

METHODOLOGY AND APPROACHES

From the analysis of methodological preferences expressed by respondents, both managerial and non-managerial, a shared perspective emerges: there is no universally "best" methodology, and the choice must necessarily depend on the specific context of the project. In particular, both groups tend to exclude the Waterfall approach, showing a strong preference for Agile methods, which are considered more flexible and better suited to the dynamic nature of software projects.

Consistent with this view, the majority of participants—regardless of role—emphasized that the methodology adopted can significantly influence the final outcome of the project. This perception aligns with the findings of the study conducted by M. Pace [9], which reported a moderate but statistically significant correlation between the methodology used and the perceived success of technological projects, with a more favorable trend toward agile methods.

The idea that methodological choice should adapt to the project context is further reinforced by the conclusions of the same study, where the author stresses the importance of greater methodological flexibility in project management. This contextual approach is also confirmed in the article by Thesing et al. [10], in which the authors argue that no universal methodological model is valid for every situation, and that the selection of an approach should be done consciously and strategically, based on the specific characteristics of each project.

In summary, these results suggest an evolution in project culture, increasingly oriented toward awareness, customization of methods, and the overcoming of rigid oppositions between traditional and agile models.

CRITICAL SUCCESS FACTORS

The analysis of the results shows an interesting convergence between the perceptions of managerial and non-managerial figures regarding critical success factors in software projects. By aggregating the responses, the five most frequently selected elements—ranked by perceived importance—are as follows:

- Clearly defined, shared, and stable requirements over time;
- Realistic planning consistent with available resources;
- Team competence;
- Customer involvement throughout the entire project lifecycle;
- Sufficient resources (human, technological, financial).

Conversely, two factors were judged as marginal in terms of their impact on project success:

- Top management support;
- Presence of iterative feedback.

This outcome reflects a practical idea of success, tied more to the operational solidity of the team, precision in planning, and clarity of shared objectives, rather than to strategic support or the formalization of feedback loops. What stands out most is the remarkable consistency between the findings of this survey and the consolidated evidence in academic literature. Each of the five identified factors directly corresponds to key concepts discussed in project management theory. This alignment is summarized in the following Table 4, which compares the most frequently mentioned success factors in the survey with their theoretical counterparts:

Emerging Factor	**Corresponding Concept in Literature**	**Explanation**
Clearly defined, shared, and stable requirement	Clear project mission	Both emphasize the need for clear, shared, and stable objectives from the start.
Realistic planning consistent with resources	Realistic planning and scheduling	Stresses planning based on concrete and sustainable data.
Team competence	Adequate team skills and leadership	Technical and relational preparation is key to successful project delivery.
Customer involvement throughout the lifecycle	Consultation and involvement of customers/users	Ongoing dialogue with stakeholders improves alignment and product adoption.
Sufficient resources	Sufficient and adequate resources	Resource adequacy is a necessary condition for project success.

Table 4 - Emerging Success Factors vs. Literature Insights

The clear alignment between field experience and established theories suggests a simple yet often overlooked truth: the foundations of project success have been known for some time, but actual compliance remains the true operational challenge. The gap lies not in knowledge, but in the ability of organizations to systematically and consistently apply these fundamental principles.

Moreover, the underestimation of factors such as top management support or structured feedback may indicate a lack of strategic awareness among many operational actors, who tend to focus on "doing things well" rather than "building the conditions to do things well." This interpretation opens up valuable reflections on how to strengthen project culture within organizations by acting on less visible but equally decisive levers.

FAILURE FACTORS

Regarding the factors perceived as most responsible for the failure of software projects, the opinions expressed by both managerial and non-managerial figures show a high level of convergence. The four most frequently mentioned elements, in order of perceived importance, are:

- Unclear or constantly changing initial requirements
- Unrealistic deadlines
- Undersized or inexperienced team
- Poor communication within the team.

Conversely, the factors considered less influential—selected by only a small minority of respondents—were:

- Conflicts between stakeholders;
- Poor budget management.

The limited importance attributed to factors such as budget management and stakeholder conflicts suggests that, in daily practice, project failures are perceived less as the result of strategic or governance choices, and more as the direct consequence of operational weaknesses. The most critical factors—unstable requirements, unrealistic deadlines, inadequately staffed teams, and communication gaps—highlight that the issues undermining success often arise within the core of project execution, affecting the team's ability to work effectively and in coordination.

Once again, the top four failure factors identified by respondents directly reflect the theoretical contributions analyzed, confirming a long-standing and consistent view of what most often jeopardizes project success. The following Table 5 shows a one-to-one comparison with the corresponding elements in the literature:

Emerging Factor	Corresponding Concept in Literature	Explanation
Unclear or constantly changing initial requirements	Undefined or constantly evolving scope (scope creep)	Both highlight the negative impact of unclear and unstable requirements, which hinder control and planning.
Unrealistic deadlines	Deadlines imposed without analysis	Arbitrary deadlines, set without a solid project foundation, jeopardize quality, team morale, and results.
Undersized or inexperienced team	Undersized or inexperienced team	The lack of qualified resources compromises operational effectiveness and project reliability.
Poor communication within the team	Inadequate communication	Fragmented or unstructured communication hinders coordination, leading to errors and inefficiencies.

Table 5 - Emerging Failure Factors vs. Literature Insights

The comparison between the findings of the survey and the main theoretical contributions reveals a striking continuity: the critical success and failure factors in software projects have not changed over time. Despite the evolution of tools, technologies, and management practices, the problems reported today by professionals match those identified in the literature even one or two decades ago.

Unstable requirements, unrealistic planning, lack of resources, communication weaknesses, and small or underqualified teams still remain the main obstacles to project success.

This data prompts an important reflection: it is not the lack of knowledge that threatens projects, but rather the systemic difficulty in translating that knowledge into operational practice.

In other words, although we have long known what the necessary conditions for success are, there still exists a gap between knowing and doing, between what theory recommends and what organizations are actually able to implement. Bridging this gap represents the true challenge for the future of project management.

ANALYSIS OF DATA ON THE LAST COMPLETED PROJECT

The inclusion of a section in the questionnaire dedicated to the last project actually completed by respondents served a clear purpose: to verify whether the critical success or failure factors identified in theory corresponded to the concrete experience lived in the field. In other words, the aim was to compare abstract perceptions with actual project realities, to understand whether professionals' beliefs were confirmed by the final outcomes of the projects in which they had participated.

The distribution between successful and failed projects showed a greater number of successful cases. However, the main focus was on the factors that respondents actually considered decisive in the success or failure of the projects they had been involved in.

For respondents in managerial or leadership roles, the factors with the greatest impact on the successful outcome of the project were:

- Clearly defined, shared, and stable requirements;
- Strong cohesion and collaboration among team members;
- Team competence;
- Sufficient resources (human, technological, financial);
- Realistic planning aligned with available resources.

These elements reveal a vision strongly focused on initial project control, team structure, and effective resource management—elements which, if well implemented, ensure stability and continuity throughout the entire project lifecycle.

Among respondents without direct managerial responsibilities, a significant alignment was also observed. The highest-rated factors were:

- Strong cohesion and collaboration among team members;
- Team competence;
- Realistic planning aligned with available resources;
- Clearly defined, shared, and stable requirements;
- Sufficient resources (human, technological, financial).

Although in a slightly different order, the same five factors emerge as decisive. The fact that operational figures prioritize team cohesion and realistic planning suggests a very practical awareness of what makes day-to-day work smooth and sustainable.

The comparison between the two profiles reveals a remarkably consistent picture. Managers and operational teams converge on the same elements as the keys to success, confirming what had already emerged during the theoretical phase of the study. This correspondence is not trivial: it demonstrates that industry professionals are well aware of what truly makes the difference in a project—and that, at least in part, they manage to translate this awareness into practice.

However, this convergence also raises an important question: if the critical success factors are well known and widely shared, why does a (moderate) portion of projects still fail to fully achieve their goals?

The answer seems to lie not in a lack of knowledge, but rather in systemic and organizational difficulties in consistently applying these principles. Requirements management, building

competent teams, realistic planning, and customer involvement are well-known elements, but they are often sacrificed due to urgent pressures, external constraints, or low organizational maturity.

In conclusion, this initial section not only confirms the validity of theoretical models, but also highlights the need to strengthen organizations' ability to systematically and coherently implement the conditions that make a project truly feasible and sustainable.

When examining the projects respondents classified as failures, some interesting divergences emerge between the perceptions of managerial and non-managerial roles, albeit within a shared framework.

For respondents in managerial or coordination roles, the main factors that negatively affected the project outcome were:

- Insufficient available resources;
- Unrealistic client expectations;
- Unrealistic deadlines;
- Unclear or constantly changing initial requirements;
- Undersized or inexperienced team.

This view highlights a dual vulnerability. On one hand, internal structural weaknesses, such as a lack of adequate resources and a weak team; on the other hand, external pressures that are difficult to control, such as unrealistic expectations and deadlines imposed by clients.

From the perspective of non-managerial figures, the most influential factors were:

- Undersized or inexperienced team;
- Unrealistic deadlines;
- Unclear or constantly changing initial requirements;
- Inadequate methodology for the context;
- Unrealistic client expectations.

In this case, alongside the organizational issues also noted by managers, an additional element stands out: the importance attributed to methodological choice. Operational roles more acutely perceive the inadequacy of the project methodology, especially when it is poorly suited to the specific context.

This observation is not isolated, but aligns with previous findings: the majority of respondents, regardless of their role, acknowledged that the methodology adopted can significantly influence the final outcome of a project.

The comparison between the two perspectives reveals both commonalities and subtle differences in how failure is interpreted. Both categories agree on the negative impact of weak teams, unrealistic

deadlines, and unstable requirements. However, only non-managerial figures explicitly emphasize methodological inadequacy, likely because they experience its operational consequences more directly.

Overall, these results suggest that despite the sector's theoretical maturity and the spread of best practices, a significant portion of projects continues to fail for known and recurring reasons. This confirms what has already been emphasized: the real gap is not between what we know and what we don't, but between what we know and what we are actually able to apply.

Overcoming this gap requires not just awareness but also structural commitment from organizations to create more resilient, flexible, and quality-oriented project environments—where quality is not just declared but truly pursued.

In the final section of the questionnaire, all participants who had completed their projects were asked to rate the extent to which key project elements were fulfilled—elements considered concrete indicators of quality and completeness. The goal was to measure not only the general perception of success or failure but also how far the fundamental outputs were actually achieved in each case.

For managerial respondents who reported a successful outcome, the four highest-rated factors were:

- Overall user satisfaction;
- Fulfillment of functional requirements;
- Compliance with technical specifications;
- Product actually in use.

This ranking reflects a holistic vision of success, integrating technical effectiveness with end-user satisfaction. It implicitly recognizes that a project can truly be considered successful only if it delivers a usable outcome, aligned with expectations, and appreciated by its users.

Interestingly, these same four elements also appeared in failed projects, albeit with lower average scores—a particularly telling result. It shows that technical quality and product usability remain central goals, even when the project as a whole does not achieve its desired outcomes. Despite failure, there is a continued effort to prioritize these fundamental aspects.

Among non-managerial roles, a similar vision emerges, with a slight variation in the order of importance. In successful projects, the most valued factors were:

- Fulfillment of functional requirements;
- Compliance with technical specifications;
- Product actually in use;
- Overall user satisfaction.

Here, the emphasis is placed more on the technical and functional adherence of the product to the initial goals, reflecting a more operational and tangible orientation. User satisfaction, while still among the top four factors, is perceived more as a consequence than a primary condition. Even for these roles, however, in failed projects, the same four factors appear—again with lower average scores—confirming that their achievement (or lack thereof) is a key lens through which project success or failure is evaluated.

These results offer an important confirmation: the perceived quality of a software project is never separated from the actual quality of the product obtained. The consistent recurrence of the same four elements—regardless of role or outcome—shows that success is assessed based on concrete, verifiable, and measurable parameters, such as user satisfaction, requirement compliance, and actual product use.

At the same time, the gap in scores between successful and failed projects, even for the same factors, suggests that recognizing the importance of an element is not enough to ensure its achievement. Intentions do not always translate into results, and this confirms that project success is not just a matter of awareness, but of coherent execution, adequate resources, and effective coordination.

In conclusion, quality—both technical and perceived—emerges as a cross-cutting, shared, and top-priority criterion for assessing the real value of a software project. As such, it should be placed at the heart of every continuous improvement strategy.

CHAPTER 4.
CONCLUSIONS

4.1. FINAL REFLECTIONS

One of the most significant findings of this study is that the success or failure of a software project cannot be attributed to a single isolated cause, but rather results from a complex and interdependent set of factors, as already highlighted by Pinto and Slevin [1] and reiterated by Pinto and Mantel [2]. The data collected through the questionnaire fully confirm this perspective: projects are the product of dynamic balances among people, processes, resources, and expectations.

Quality as a Fixed Point, Beyond Constraints

The centrality of quality strongly emerges as the essential criterion for evaluating the success of a software project, surpassing traditional constraints such as time and cost. Both managers and operators agree in considering the quality of the final product the most important factor—consistent with the insights of Agarwal and Rathod [5], who emphasize that success is mainly defined by the ability to deliver a sustainable and useful product, even at the cost of sacrificing budget and deadlines. Quality thus appears not only as a technical metric, but as a universally shared anchor point, encompassing the fulfillment of functional requirements and technical specifications, actual use of the product, and end-user satisfaction. This evidence suggests that quality is not a luxury reserved for well-managed projects, but a priority even in the most critical situations, underlining its central role in software project management and evaluation.

Methodology: Adaptability is Key

The analysis of data on work methodologies shows a broad consensus that no methodology is universally superior, but that the choice must be contextual and deliberate. The responses confirm that, although agile methods are currently preferred, what really matters is the ability to adapt the approach to the nature of the project and the organizational context. As shown in the studies by Pace [9] and by Thesing, Feldmann, and Burchardt [10], methodological flexibility is a critical success factor, especially in the complex and dynamic world of software development.

Critical Factors: Consistency with the Literature

Another significant result concerns the convergence between professionals' opinions and what is stated in the scientific literature. The most frequently cited critical success factors—clear requirements, realistic planning, team competencies, adequate resources, and customer involvement—correspond exactly to those identified by numerous academic studies. This finding

suggests that the conditions for success are well-known and widely shared, but often difficult to implement systematically. The gap does not lie in lack of knowledge, but in the organizational ability to consistently and coherently apply these established principles.

Likewise, the most recurring failure factors—unstable requirements, unrealistic deadlines, weak teams, and inadequate communication—reflect weaknesses that have been documented for years. Despite the evolution of tools and methodologies, the most frequent problems are still rooted in the operational dimension of the project, indicating the need to strengthen prevention, monitoring, and support mechanisms at the execution level.

Concrete Experience and Perception: The Test of the Last Project

A key section of the survey asked respondents to evaluate their most recently completed project, with the goal of comparing theoretical opinions with real-life experience. Again, the responses confirmed a strong coherence: the same factors identified as critical in theory turned out to be the most influential in actual project outcomes, whether successful or not. Managerial and operational roles showed minimal differences in their perception of relevant factors, demonstrating a shared awareness across all team levels.

However, the fact that these same factors also appeared in failed projects—albeit with lower average scores—suggests that intention alone is not enough: coherent execution, adequate resources, and ongoing coordination are needed to turn conditions into results. The real challenge lies in the shift from "knowing what matters" to "knowing how to implement it".

Final Thought

In summary, this research clearly shows that the foundations of success and failure in software projects are well understood by professionals, but their implementation remains an ongoing challenge. The consistency of the results, their alignment with the literature, and the convergence across different organizational roles indicate that the levers of project management are well-known and shared. However, the effectiveness of a project still depends on organizations' ability to translate awareness into practice and models into coherent operational behaviors.

The future of project management does not require new theories, but new ways of applying what we already know: more focus on execution, more investment in communication and team competencies, and a continuous commitment to creating contexts that prioritize quality as a primary objective, rather than a secondary outcome.

CHAPTER 5.
FUTURE DEVELOPMENTS

5.1 PROPOSALS FOR FUTURE INSIGHTS

The results of this research have confirmed the complexity of project management in the software domain and, at the same time, highlighted a significant consistency between operational perceptions and well-established literature. However, this very convergence raises crucial questions: if the critical success and failure factors have been known for decades, why do instability, delays, and failures continue to occur?

To answer this question, it is necessary to go beyond the descriptive approach and move toward more analytical, longitudinal, and multivariate studies capable of investigating the interactions between factors, the deep systemic causes, and the cultural and organizational dynamics that hinder the effective application of best practices.

Longitudinal Studies and Temporal Analyses

A first development direction concerns the implementation of longitudinal studies, which would allow the observation of the evolution of the same project or portfolio of projects over time. These studies would make it possible to understand:

- How perceptions of critical factors change throughout the project lifecycle;
- What patterns of failure or success recur systemically;
- How corrective actions introduced during the project influence the final outcomes.

This type of approach would be particularly useful in agile scenarios, where feedback is frequent and planning is iterative.

Integration of "Hard" and "Soft" Factors

This research focused primarily on objective and measurable factors (requirements, resources, planning). However, an underexplored aspect is the effect of relational and psychological dynamics on project success. Future studies could investigate:

- The role of emotional leadership and conflict management within the team;
- The impact of motivation, stress, and cognitive load on performance;
- The relationships between team engagement and final customer satisfaction.

This would involve combining quantitative tools with qualitative methods (e.g., interviews and organizational climate analysis).

Predictive Modeling of Project Risk

Finally, the data collected through surveys like the one proposed here could form the basis for the development of predictive models of project failure risk, using advanced artificial intelligence techniques such as:

- Supervised machine learning, for example to automatically classify projects based on risk level;
- Bayesian analysis, to model contextual factor interactions and estimate the probability of failure;
- Dynamic risk scorecards, directly integrable into project management systems to support timely and informed decisions.

The goal is to evolve from a purely descriptive approach to one that is predictive and preventive, capable of guiding resource allocation and project planning in a smarter, more efficient, and data-driven way.

These technologies pave the way for a new generation of tools and methodologies, where artificial intelligence acts as a strategic ally to the project manager, helping to close the persistent gap between what theory suggests and what organizations actually implement.

Final Consideration

In light of the results obtained, it is clear that the future of project management cannot disregard ongoing technological evolution—but it must not be reduced to it either. Artificial intelligence offers enormous potential, but its effectiveness will depend on the ability of organizations to integrate it with sensitivity, awareness, and a value-oriented approach.

The theoretical foundations are solid, and field experience is rich. Now is the time to invest in the intelligent translation of knowledge into systems and processes that truly support those working on projects, reducing the gap between what we know and what we accomplish. A gap that, if tackled with the right tools, can finally be closed.

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